```sql
-- DDL Commands

--1. Create Command
create database SchoolDb
use SchoolDb;

-- 2. Drop Command
Drop database SchoolDb;

create table students(
id int primary key ,
name varchar(100)
);

-- DML Commands
-- 1. Insert Command
insert into students values(1,'Rohit');
insert into students values(2,'Devansh');
insert into students values(3,'Bhanu');

-- 2. Update Command
-- Syntax -> update tablename set columnName = value condition

update students set name = 'krishna' where id=2;

-- 3. Delete Command
-- Syntax -> Delete from tableName where Condition
delete from students where id=2;

-- DQL Commands

-- 1. Select Command
-- Syntax -> Select * from tableName
select * from students
select name from students where id =3

-- DCL Commands
-- 1. Grant Command
-- Syntax -> grant on to ;
/*
permission → What access you are giving (SELECT, INSERT, UPDATE, DELETE, etc.)
object → Table, View, Stored Procedure, Database, Schema, etc.
principal → Login, User, or Role
*/
```

```sql
grant select on dbo.students to User1;

--2. Revoke Command
-- Syntax -> revoke on to ;
revoke select on dbo.students to User1;

-- TCL Commands
--1. Commit

begin tran;
insert into students values(4,'Amit');
update students set name = 'Rohit Kumar' where id=1 ;
commit;

--2. Rollback

begin tran;
insert into students values(6,'Raj');
delete from students where id=1;
rollback;

-- 3. Savepoint

BEGIN TRAN;
INSERT INTO students VALUES (7, 'Pooja');
SAVE TRANSACTION A;
UPDATE students SET name = 'Bhanu Kumar' WHERE id = 3;
ROLLBACK TRANSACTION A; -- Only rolls back changes after savepoint A
COMMIT; -- Saves remaining changes

-- Joins-----
CREATE TABLE Patient (
patientId INT PRIMARY KEY IDENTITY(1,1),
Name VARCHAR(100),
DOB DATE,
bloodGroup VARCHAR(10),
phone BIGINT UNIQUE
);

INSERT INTO Patient (Name, DOB, bloodGroup, phone) VALUES
('Rohit', '1998-05-12', 'B+', 9876543210),
('Amit', '2000-08-20', 'O+', 9876543211),
('Neha', '1997-03-15', 'A+', 9876543212),
```

```sql
('Rahul', '1995-11-30', 'AB+', 9876543213);

CREATE TABLE Appointment (
appointmentId INT PRIMARY KEY IDENTITY(1,1),
patientId INT,
appointmentDate DATE,
doctorName VARCHAR(100),
status VARCHAR(20),
FOREIGN KEY (patientId) REFERENCES Patient(patientId)
);

INSERT INTO Appointment (patientId, appointmentDate, doctorName, status) VALUES
(1, '2024-02-10', 'Dr. Sharma', 'Scheduled'),
(2, '2024-02-11', 'Dr. Mehta', 'Completed'),
(1, '2024-02-12', 'Dr. Verma', 'Cancelled'),
(3, '2024-02-13', 'Dr. Sharma', 'Scheduled');

-- Inner Join----
select p.patientId, p.Name, a.appointmentDate, a.doctorName, a.status from Patient p inner join
Appointment a on p.patientId=a.patientId

-- Left Join---
SELECT p.patientId, p.Name, a.appointmentDate, a.doctorName, a.status
FROM Patient p
LEFT JOIN Appointment a
ON p.patientId = a.patientId;

-- Right Join ----
SELECT p.patientId, p.Name, a.appointmentDate, a.doctorName, a.status
FROM Patient p
RIGHT JOIN Appointment a
ON p.patientId = a.patientId;

-- Full Outer Join
SELECT
p.patientId,
p.Name,
a.appointmentDate,
a.doctorName,
a.status
FROM Patient p
FULL OUTER JOIN Appointment a
```

```
ON p.patientId = a.patientId;
```